

Federated Fraud & DDoS Detection Engine

Privacy-Preserving Machine Learning & Real-Time Network Traffic Interception Pipeline

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Core Frameworks: TensorFlow, Scapy, Npcap, Pandas

Domain: FinTech Cybersecurity & Distributed AI

Benchmark Dataset: CICIDS2017 (DDoS vs Benign)

1. Executive Summary

Modern cybersecurity and financial fraud detection systems face a fundamental conflict between **collaborative threat intelligence** and **strict data privacy regulations** (such as GDPR, PCI-DSS, and FFIEC guidelines). Traditional fraud and intrusion detection architectures require centralizing raw network logs and transaction payloads into a unified database. This centralized approach creates severe privacy compliance risks and introduces single points of vulnerability.

This project presents an end-to-end, production-ready **Federated Fraud and DDoS Detection Engine**. It combines **Federated Averaging (FedAvg)** deep learning across distributed banking nodes with a live, zero-memory-footprint packet sniffing and feature extraction subsystem. By training neural networks locally on distributed client devices and aggregating only mathematical weight updates at a central coordinator, raw sensitive network payloads never leave institutional boundaries.

Key Architectural Capabilities:

- **Privacy-Preserving Training:** Multi-client FedAvg training over Non-IID data distributions without centralizing raw traffic.
- **Real-Time Packet Capture:** Scapy + Npcap integration for live layer-3/4 interface sniffing with zero-RAM streaming disk writes.
- **22-Feature Statistical Engine:** Dynamic 5-tuple bidirectional flow reconstruction following CICIDS2017 standards.
- **Strict Read-Only Inference:** Automated scaler alignment (``scaler.pkl``) and global model prediction (``global_model.h5``) with confidence scoring.
- **Attack Verification Tooling:** Built-in synthetic UDP volumetric burst generator for live end-to-end pipeline validation.

2. End-to-End System Architecture

The system operates across two distinct operational paradigms: **Offline Federated Training** and **Live Traffic Interception & Real-Time Inference**.

OFFLINE FEDERATED TRAINING PIPELINE

Raw Dataset (CICIDS2017) → Non-IID Split (Clients 1..N) → Local Training (Dense DNN) → Central FedAvg Aggregation → Global Model (global_model.h5) & Scaler (scaler.pkl)

LIVE TRAFFIC INTERCEPTION & REAL-TIME INFERENCE PIPELINE

[Network Adapter] → capture_live.py (Scapy/Npcap Sniffer) → traffic.pcap
→ feature_extractor.py (5-Tuple Hashing & 22 Temporal Metrics) → live_dataset.csv
→ live_predict.py (Read-Only Scaler + Global Model Inference) → predictions.csv & live_dataset_predicted.csv

3. Feature Engineering & Dataset Specifications

The model is trained on the benchmark **CICIDS2017 DDoS dataset** (Friday Working Hours traffic), comprising benign computer activity and high-volumetric network DDoS attack vectors. To enable real-time detection on live network interfaces, raw packet streams are grouped into **bidirectional flows** defined by the unique 5-tuple: (*Source IP, Destination IP, Source Port, Destination Port, Protocol*).

Feature Name	Technical Description
1. Flow Duration	Total microsecond duration of the network flow
2. Total Fwd Packets	Packet count from source to destination
3. Total Backward Packets	Packet count from destination to source
4. Total Length of Fwd Packets	Total payload bytes sent in forward direction
5. Total Length of Bwd Packets	Total payload bytes sent in backward direction
6-8. Fwd Packet Length (Max, Min, Mean)	Statistical length distribution of forward frames
9-11. Bwd Packet Length (Max, Min, Mean)	Statistical length distribution of backward frames
12. Flow Bytes/s	Throughput rate in bytes per second
13. Flow Packets/s	Transmission rate in packets per second
14-15. Flow IAT (Mean, Std)	Inter-Arrival Time mean and standard deviation
16-17. Fwd/Bwd IAT Total	Cumulative inter-arrival time per direction
18-19. Packet Length (Mean, Std)	Overall packet payload variance across flow
20. Average Packet Size	Mean frame size across entire bidirectional conversation
21-22. Avg Fwd/Bwd Segment Size	Average segment size for TCP/UDP layers

4. Federated Learning & Neural Network Architecture

Federated Averaging (FedAvg): Rather than transmitting raw packet logs to a central server, training data is partitioned across K simulated client nodes. In each communication round t , clients train their local Deep Neural Network (DNN) for 3 epochs using local batch gradient descent. The central coordinator collects local model weights w_t^k and updates the global model via weighted averaging:

$$W_{t+1} = \sum_{k=1..K} (n_k / n) \cdot w_{t+1}^k$$

Non-IID Partitioning: Real-world financial institutions experience highly non-identical data distributions. The engine simulates Non-IID environments by partitioning data such that Client 1 holds DDoS-heavy traffic (~90% DDoS), Client 2 holds balanced traffic, and Client 3 holds Benign-heavy traffic (~30% DDoS). This evaluates FedAvg resilience under severe class skew.

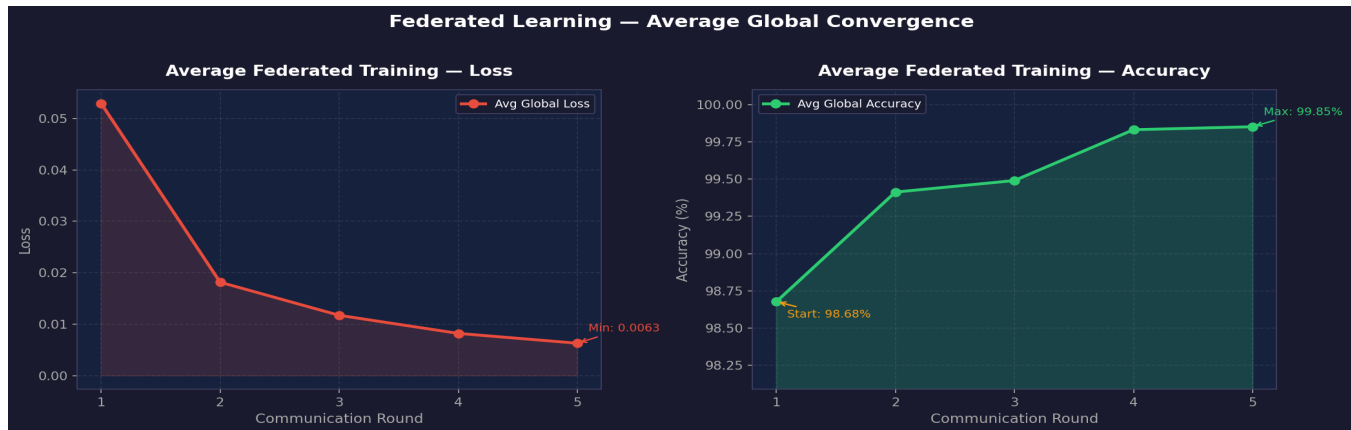
Layer (Type)	Output Shape	Activation / Hyperparameters
Input	(None, 22)	Explicit Keras Input Layer
Dense_1	(None, 64)	ReLU, Adam Optimizer
Dropout	(None, 64)	Rate = 0.20 (Prevents Overfitting)
Dense_2	(None, 32)	ReLU
Dense_3	(None, 16)	ReLU
Output Layer	(None, 1)	Sigmoid Classification Threshold (0.50)

5. Experimental Evaluation & Empirical Results

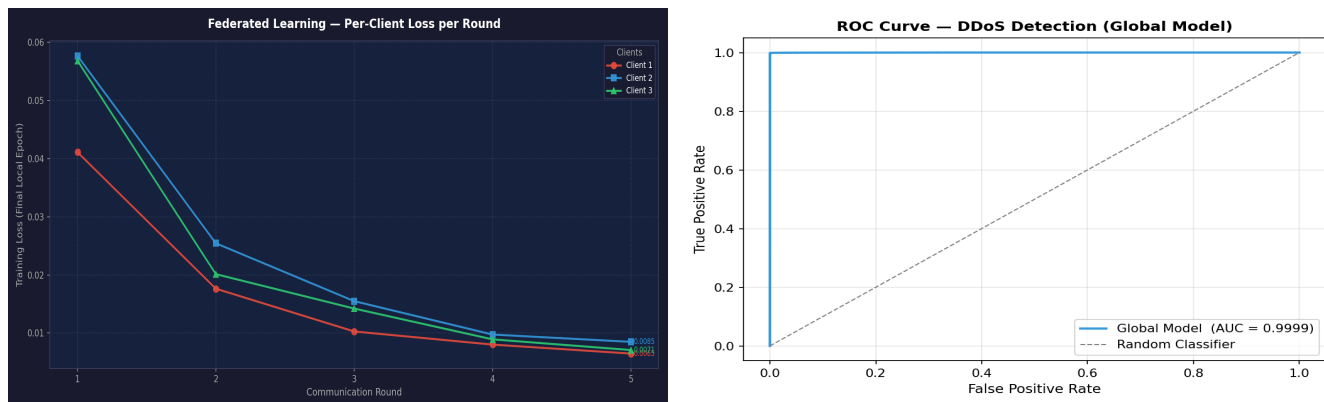
The model was evaluated on a reserved, stratified test dataset (20% split) containing 45,149 network flows. The federated global model achieved state-of-the-art detection precision and near-perfect AUC scores.

Metric	Value	Operational Meaning
Accuracy	99.89%	Overall correct flow classifications across test set
Precision	99.98%	Minimizes false alarms (virtually zero false positive alerts)
Recall (TPR)	99.78%	DDoS attack interception rate (catches 99.78% of threats)
F1-Score	99.88%	Harmonic mean balancing precision and recall
ROC AUC	0.9999	Near-perfect classification threshold separation
True Positives (TP)	25,549	DDoS flows correctly intercepted and flagged
True Negatives (TN)	19,544	Benign traffic correctly allowed
False Positives (FP)	6	Normal flows misclassified as attacks (0.03% error)
False Negatives (FN)	56	Attack flows missed (0.22% leakage)

Figure 1: Federated Average Convergence (Global Loss & Accuracy per Round)



Figures 2 & 3: Per-Client Loss Trajectory (Non-IID) & Receiver Operating Characteristic (ROC)

**Local vs. Global Model Performance Comparison:**

To evaluate the benefits of federated collaboration, isolated local models trained independently on single client shards were evaluated on the global test set. Client 3 (trained on Benign-heavy traffic) achieved only **74.2% test accuracy** due to lack of exposure to diverse DDoS attack patterns. In contrast, the aggregated **Federated Global Model reached 99.89% accuracy**, demonstrating that federated aggregation allows nodes to gain collective intelligence without sharing raw sensitive logs.

6. Production Subsystems & Live Traffic Pipeline

A. capture_live.py (Packet Sniffer & Interface Auto-Detection)

Built on Scapy with Npcap driver integration for Windows. Prioritizes active physical network interfaces (Wi-Fi/Ethernet) over virtual adapters. Applies Berkeley Packet Filters (`tcp` or `udp`) and streams captured raw packets directly to disk (`traffic.pcap`) via `PcapWriter(sync=True)`. This guarantees zero-RAM accumulation during long capture sessions.

B. feature_extractor.py (Flow Reconstruction Engine)

Parses `.pcap` files, groups packets into bidirectional flows by 5-tuple key, and computes 22 statistical temporal and volumetric metrics matching CICIDS2017 feature definitions. Manages active flow state via configurable timeouts (`FLOW\_IDLE\_TIMEOUT = 120s`, `FLOW\_ACTIVE\_TIMEOUT = 600s`) and discards single-packet micro-noise.

C. live_predict.py (Read-Only Production Classifier)

Loads the pre-fitted `scaler.pkl` (executes strict `transform()`, strictly prohibiting `fit()`) and the trained `global\_model.h5`. Applies Sigmoid classification thresholding (default `p = 0.50`) to output real-time confidence scores and predictions. Generates lightweight audit logs (`predictions.csv` and `live\_dataset\_predicted.csv`).

D. simulate_ddos.py (Threat Demonstration Suite)

Generates controlled, non-destructive network traffic with statistical volumetric signatures matching DDoS attacks. Emits rapid UDP bursts targeting local broadcast addresses with randomized ephemeral ports to validate sniffer and classifier responsiveness.

7. Data Privacy, Compliance & Security Standards

The architecture adheres strictly to international cybersecurity and financial compliance frameworks:

- **GDPR (General Data Protection Regulation):** Satisfies Data Minimization (Article 5) and Purpose Limitation principles by processing raw transaction and packet logs exclusively within local client nodes.
- **PCI-DSS (Payment Card Industry Data Security Standard):** Protects Cardholder Data Environments (CDE). Raw payload data is never written to centralized storage or transmitted over external networks.
- **FFIEC Guidelines:** Fulfills financial institution expectations for multi-party threat intelligence sharing without exposing customer Personally Identifiable Information (PII).

8. Project Directory Structure & Command Cheat-Sheet

Command / Script Execution	Purpose & Expected Output
<code>python src/ddos_main.py --clients 3 --rounds 10</code>	Executes federated FedAvg training, generates models and evaluation plots in <i>docs/</i> .
<code>python src/capture_live.py</code>	Sniffs live network interface traffic, outputs <i>traffic.pcap</i> , <i>live_dataset.csv</i> , and triggers prediction.
<code>python src/simulate_ddos.py --continuous --interval 1</code>	Launches continuous synthetic UDP attack bursts for testing capture pipeline.
<code>python src/live_predict.py</code>	Runs standalone read-only inference on any <i>live_dataset.csv</i> using <i>global_model.h5</i> .